॥ वसुधैव कुटुम्बकम् ॥

SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

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Subject: Operating Systems Lab	**Sem:** III	**AY:** 2026-27
EXPERIMENT No : 2		
TITLE : Basic Shell Commands		**DoP : 14-07-2026**

Aim: Demonstrate the use of Basic Shell Commands

Learning Outcome: 1. To understand the shell commands
2. To demonstrate the shell commands

Hardware/Software:

Theory:

The shell acts as an interface between the user and the OS services. It accepts human-readable commands from the user and converts them into a language that the kernel can understand. The shell can be accessed using the terminal.

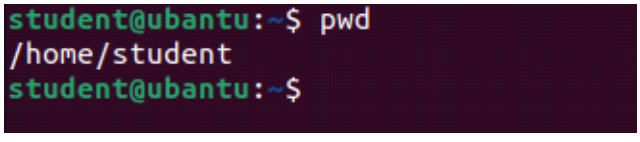
Some of the basic shell commands are:

Command	***Description***
pwd	Print the current working directory
ls	List the files and directories within the current working directory
cd	Change the current working directory
touch	Create, change or modify file timestamps – create a file without any content
mkdir	Create a new directory (folder)
cat	Create, view or concatenate files
rm	Remove files
bc	Command line basic calculator to do basic mathematics calculations
tty	Print the filename of the terminal connected to the standard input
who	Print all usernames currently logged in
man	Display the user manual of any command

date	Display the date and time of the system, or may be used to set the date and time when used as the root user.
history	Display the list of commands previously executed
ps	Process Status – List currently running processes and their PIDs with some other information
time	Print summary of real-time, user time and system CPU time spend by executing a command when it terminates. 'real' time is the time elapsed wall clock time taken by a command to get executed, while 'user' and 'sys' time are the number of CPU seconds that command uses in user and kernel mode respectively.
head	Print the top N number of data of the given input. By default, it prints the first 10 lines of the specified files.
tail	Print the last N number of data of the given input. By default, it prints the last 10 lines of the specified files.
cp	Copy files or groups of files or directories. It creates an exact image of a file on a disk with a different file name. It requires at least two filenames as arguments.
sort	Sort text files.
wc	Print byte count, word count, and line count.
chmod	Change Mode - Change the access permissions of files and directories.
rmdir	Remove directories/folders.
cut	Used to remove sections or columns from each line of a file and display the selected parts of the content.
top	Displays real-time information about running processes, CPU usage, memory usage, and system performance.
gedit	A graphical text editor in Linux used to create, view, and edit text files.
cd ..	Moves the user one level up to the parent directory of the current working directory.

Output Screenshots:

1. pwd



2. ls

```
student@ubantu:~$ ls
 AAH       Demo2.sh   Desktop    Downloads   new_f     Pictures   snap       Videos
 Demo1.sh  Demo3.sh   Documents  Music      'OS DSY'   Public     Templates
student@ubantu:~$
```

3. touch

```
student@ubantu:~/OS$ touch sample2.txt
student@ubantu:~/OS$ ls
sample2.txt  sample.txt
student@ubantu:~/OS$
```

4. mkdir

```
student@ubantu:~$ mkdir new_folder
student@ubantu:~$ ls
 AAH        Demo2.sh   Desktop     Downloads   new_f       'OS DSY'    Public    Templates
 Demo1.sh   Demo3.sh   Documents   Music       new_folder   Pictures   snap      Videos
student@ubantu:~$
```

5. cat

```
student@ubantu:~/OS$ cat sample.txt
hello
student@ubantu:~/OS$ cat>sample.txt
hey
welcome to SIT ,pune
student@ubantu:~/OS$ cat sample.txt
hey
welcome to SIT ,pune
student@ubantu:~/OS$ cat>>sample.txt
this college in india.
student@ubantu:~/OS$ cat sample.txt
hey
welcome to SIT ,pune
this college in india.
student@ubantu:~/OS$
```

6. rm

```
student@ubantu:~/OS$ rm sample2.txt
student@ubantu:~/OS$ ls
sample.txt
student@ubantu:~/OS$
```

7. bc

```
student@ubantu:~/OS$ bc
bc 1.07.1
Copyright 1991-1994, 1997, 1998, 2000, 2004, 2006, 2008, 2012-2017 Free Software Foundation, Inc.
This is free software with ABSOLUTELY NO WARRANTY.
For details type `warranty'.
10+1
11
100-20
80
12*2
24
100/2
50
15%3
0
10>7
1
5<15
1
6>10
0
quit
student@ubantu:~/OS$
```

8. tty

```
student@ubantu:~/OS$ tty
/dev/pts/0
```

9. who

```
student@ubantu:~/OS$ who
student  tty2         2026-07-14 09:07 (tty2)
student@ubantu:~/OS$
```

10. man

```
student@ubantu:~/OS$ man
What manual page do you want?
For example, try 'man man'.
student@ubantu:~/OS$ man bc
```



11. date

```
student@ubantu:~/OS$ date
Tuesday 14 July 2026 09:54:58 AM IST
student@ubantu:~/OS$ date +%h
Jul
student@ubantu:~/OS$ date +%m
07
student@ubantu:~/OS$ date +%y
26
student@ubantu:~/OS$ date +%Y
2026
student@ubantu:~/OS$ date +%p
AM
student@ubantu:~/OS$ date +%d
14
student@ubantu:~/OS$ date +%a
Tue
student@ubantu:~/OS$ date +%A
Tuesday
student@ubantu:~/OS$ date +%B
July
student@ubantu:~/OS$
```

12. history

```
student@ubantu:~/OS$ history
    1  gedit Demo.h
    2  gedit Demo1.sh
    3  bash Demo1.sh
    4  gedit Demo1.sh
    5  init 0
    6  ls
    7  cd Desktop
    8  gedit ESE1.sh
    9  bash ESE1.sh
   10  gedit ESE1.sh
   11  bash ESE1.sh
   12  gedit ESE1.sh
   13  bash ESE1.sh
   14  gedit ESE1.sh
   15  bash ESE1.sh
   16  gedit ESE2.sh
   17  bash ESE2.sh
   18  gedit ESE2.sh
   19  bash ESE2.sh
   20  gedit ESE2.sh
   21  bash ESE2.sh
   22  gedit ESE2.sh
   23  bash ESE2.sh
   24  gedit ESE2.sh
   25  bash ESE2.sh
   26  gedit ESE2.sh
   27  bash ESE2.sh
   28  gedit ESE2.sh
   29  bash ESE2.sh
   30  cd Desktop
   31  gedit ESE1.sh
   32  bash ESE1.sh
   33  gedit ESE2.sh
   34  bash ESE2.sh
   35  gedit ESE2.sh
   36  bash ESE2.sh
   37  gedit ESE2.sh
   38  bash ESE2.sh
   39  gedit ESE1.sh
```

13. ps

```
student@ubantu:~/OS$ ps
    PID TTY          TIME CMD
   3453 pts/0    00:00:00 bash
  54355 pts/0    00:00:00 ps
student@ubantu:~/OS$
```

14. time

```
student@ubantu:~/OS$ time

real    0m0.000s
user    0m0.000s
sys     0m0.000s
student@ubantu:~/OS$
```

15. head

```
student@ubantu:~/OS$ head sample.txt
1 hello world
ABC DEF GHI
test123 sample
@#Special!Line
small letters only
BIG LETTERS ONLY
9876543210
Mix123OfChars
$Money%Value&
abcdefghijklmnopqrstuvwxyz
student@ubantu:~/OS$ head -5 sample.txt
1 hello world
ABC DEF GHI
test123 sample
@#Special!Line
small letters only
student@ubantu:~/OS$
```

```
student@ubantu:~/OS$ head -c 12 sample.txt
1 hello worlstudent@ubantu:~/OS$
```

16. tail

```
student@ubantu:~/OS$ tail sample.txt
OpenAI ChatGPT 2026
sample_text_file
123abcABC
Small big MIX
[Special] {Chars}
Password@123
File_Name#01
END OF SAMPLE
123456789
42643654
student@ubantu:~/OS$ tail -3 sample.txt
END OF SAMPLE
123456789
42643654
student@ubantu:~/OS$
```

```
student@ubantu:~/OS$ tail -c 12 sample.txt
89
42643654
```

17. cp

```
student@ubantu:~/OS$ touch sample2.txt
student@ubantu:~/OS$ cat sample.txt
hey
welcome to SIT ,pune
this college in india.
student@ubantu:~/OS$ cp sample.txt sample2.txt
student@ubantu:~/OS$ cat sample2.txt
hey
welcome to SIT ,pune
this college in india.
student@ubantu:~/OS$
```

18. sort

```
student@ubantu:~/OS$ sort sample.txt
!@#$%^&*()
$Money%Value&
123456789
123abcABC
1 hello world
42643654
9876543210
ABC DEF GHI
abcdefghijklmnopqrstuvwxyz
ABCDEFGHIJKLMNOPQRSTUVWXYZ
abcXYZ789
BIG LETTERS ONLY
Data_Test_01
END OF SAMPLE
File_Name#01
Hello@123
LineNumber25
Mix123OfChars
OpenAI ChatGPT 2026
Password@123
sample_text_file
Small big MIX
small letters only
[Special] {Chars}
@#Special!Line
test123 sample
WELCOME TO TXT FILE
student@ubantu:~/OS$
```

```
student@ubantu:~/OS$ sort -r sample.txt
WELCOME TO TXT FILE
test123 sample
@#Special!Line
[Special] {Chars}
small letters only
Small big MIX
sample_text_file
Password@123
OpenAI ChatGPT 2026
Mix123OfChars
LineNumber25
Hello@123
File_Name#01
END OF SAMPLE
Data_Test_01
BIG LETTERS ONLY
abcXYZ789
ABCDEFGHIJKLMNOPQRSTUVWXYZ
abcdefghijklmnopqrstuvwxyz
ABC DEF GHI
9876543210
42643654
1 hello world
123abcABC
123456789
$Money%Value&
!@#$%^&*()
student@ubantu:~/OS$
```

```
student@ubantu:~/OS$ sort -n sample.txt
!@#$%^&*()
$Money%Value&
ABC DEF GHI
abcdefghijklmnopqrstuvwxyz
ABCDEFGHIJKLMNOPQRSTUVWXYZ
abcXYZ789
BIG LETTERS ONLY
Data_Test_01
END OF SAMPLE
File_Name#01
Hello@123
LineNumber25
Mix123OfChars
OpenAI ChatGPT 2026
Password@123
sample_text_file
Small big MIX
small letters only
[Special] {Chars}
@#Special!Line
test123 sample
WELCOME TO TXT FILE
1 hello world
123abcABC
42643654
123456789
9876543210
```

19. wc

```
student@ubantu:~/OS$ wc sample.txt
 27  46 400 sample.txt
student@ubantu:~/OS$ wc -l sample.txt
27 sample.txt
student@ubantu:~/OS$ wc -w sample.txt
46 sample.txt
student@ubantu:~/OS$ wc -c sample.txt
400 sample.txt
student@ubantu:~/OS$ wc -m sample.txt
400 sample.txt
student@ubantu:~/OS$
```

20. chmod

```
student@ubantu:~/OS$ chmod +w sample.txt
student@ubantu:~/OS$ chmod +r sample.txt
student@ubantu:~/OS$ chmod +x sample.txt
```

21. rmdir

```
student@ubantu:~$ rmdir new_folder
student@ubantu:~$ ls
 AAH         Demo2.sh   Desktop     Downloads   new_f     Pictures   snap        Videos
 Demo1.sh    Demo3.sh   Documents   Music      'OS DSY'   Public     Templates
student@ubantu:~$
```

22. cut

```
student@ubantu:~/OS$ cut -b 1,2,3,4 sample.txt
1 he
ABC
test
@#Sp
smal
BIG
9876
Mix1
$Mon
abcd
ABCD
Hell
Data
WELC
abcX
!@#$
Line
Open
samp
123a
Smal
[Spe
Pass
File
END
1234
4264
```

```
student@ubantu:~/OS$ cut -b 1-7,9-10 sample.txt
1 hellowo
ABC DEFGH
test123sa
@#Specil!
small ltt
BIG LETER
987654310
Mix123OCh
$Money%al
abcdefgij
GIJ
Help
@13
Data_Tet_
WELCOMETO
abcXYZ79
!@#$%^&()
LineNumer
OpenAI ha
sample_ex
123abcAC
Small bg
[Specia]
Passwor@1
File_Nae#
END OF AM
12345679
4264365
```

```
student@ubantu:~/OS$ cut -b 1- sample.txt
1 hello world
ABC DEF GHI
test123 sample
@#Special!Line
small letters only
BIG LETTERS ONLY
9876543210
Mix123OfChars
$Money%Value&
abcdefghijklmnopqrstuvwxyz
ABCDEFGHIJKLMNOPQRSTUVWXYZ
Hello@123
Data_Test_01
WELCOME TO TXT FILE
abcXYZ789
!@#$%^&*()
LineNumber25
OpenAI ChatGPT 2026
sample_text_file
123abcABC
Small big MIX
[Special] {Chars}
Password@123
File_Name#01
END OF SAMPLE
123456789
42643654
```

```
student@ubantu:~/OS$ cut -b 2-  sample.txt
 hello world
BC DEF GHI
est123 sample
#Special!Line
mall letters only
IG LETTERS ONLY
876543210
ix123OfChars
Money%Value&
bcdefghijklmnopqrstuvwxyz
BCDEFGHIJKLMNOPQRSTUVWXYZ
ello@123
ata_Test_01
ELCOME TO TXT FILE
bcXYZ789
@#$%^&*()
ineNumber25
penAI ChatGPT 2026
ample_text_file
23abcABC
mall big MIX
Special] {Chars}
assword@123
ile_Name#01
ND OF SAMPLE
23456789
2643654
```

```
student@ubantu:~/OS$ cut -c 2,3,4,5 sample.txt
 hel
BC D
est1
#Spe
mall
IG L
8765
ix12
Mone
bcde
BCDE
ello
ata_
ELCO
bcXY
@#$%
ineN
penA
ampl
23ab
mall
Spec
assw
ile_
ND O
2345
2643
```

```
student@ubantu:~/OS$ cut -c 2-3 sample.txt
 h
BC
es
#S
ma
IG
87
ix
Mo
bc
BC
el
at
EL
bc
@#
in
pe
am
23
ma
Sp
as
il
ND
23
26
```

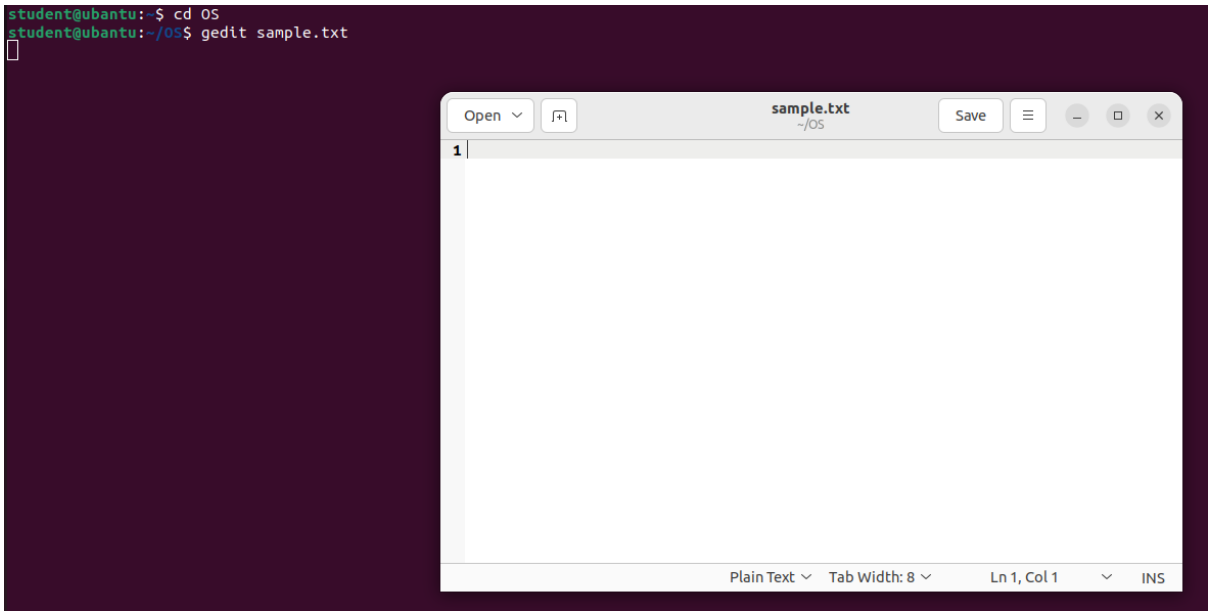
23. top

```
student@ubantu:~/OS$ top
```

```
top - 09:47:01 up 39 min,  1 user,  load average: 0.99, 1.03, 0.69
Tasks: 197 total,   3 running, 193 sleeping,   0 stopped,   1 zombie
%Cpu(s):  7.3 us, 10.0 sy,  2.4 ni, 68.0 id,  5.7 wa,  0.0 hi,  6.5 si,  0.0 st
MiB Mem :   3907.0 total,    136.8 free,   1084.6 used,   2685.5 buff/cache
MiB Swap:   3898.0 total,   3897.5 free,      0.5 used.   2497.6 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
  56970 root      20   0   66044  59464   2816 R  10.6   1.5   0:00.32 dpkg
   3870 root      20   0  437756 268932 166016 S  10.0   6.7   4:24.91 unattended-upgr
   1458 student   20   0 4833708 347752 142084 S   5.6   8.7   1:17.04 gnome-shell
  56893 root      20   0  437756 171412  68352 S   4.0   4.3   0:00.12 unattended-upgr
     71 root       0 -20       0      0      0 I   2.0   0.0   0:01.99 kworker/2:1H-kblockd
  56978 root      20   0   19052  10624   2048 S   2.0   0.3   0:00.06 dpkg-deb
     10 root      20   0       0      0      0 I   1.3   0.0   0:02.42 kworker/u6:0-ext4-rsv-conversion
   3353 student   20   0  565080  54720  42108 S   0.7   1.4   0:04.09 gnome-terminal-
     16 root      20   0       0      0      0 I   0.3   0.0   0:01.48 rcu_preempt
     53 root       0 -20       0      0      0 I   0.3   0.0   0:02.34 kworker/0:1H-kblockd
    103 root       0 -20       0      0      0 I   0.3   0.0   0:01.84 kworker/1:1H-kblockd
    165 root      20   0       0      0      0 S   0.3   0.0   0:02.00 jbd2/sda3-8
   1969 student   20   0 2862988  61804  44824 S   0.3   1.5   0:01.87 gjs
  10477 root      20   0       0      0      0 I   0.3   0.0   0:01.44 kworker/u6:3-ext4-rsv-conversion
  10997 systemd+  20   0   14868   6912   6144 S   0.3   0.2   0:00.38 systemd-oomd
      1 root      20   0   20436  12928   8320 S   0.0   0.3   0:02.90 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.01 kthreadd
      3 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_gp
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_par_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 slub_flushwq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 netns
     11 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 mm_percpu_wq
     12 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
     13 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
     14 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
     15 root      20   0       0      0      0 S   0.0   0.0   0:00.46 ksoftirqd/0
     17 root      rt   0       0      0      0 S   0.0   0.0   0:00.06 migration/0
     18 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/0
     19 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
     20 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
     21 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/1
     22 root      rt   0       0      0      0 S   0.0   0.0   0:00.39 migration/1
     23 root      20   0       0      0      0 S   0.0   0.0   0:00.91 ksoftirqd/1
```

24. gedit




```
student@ubantu:~$ cd OS
student@ubantu:~/OS$ gedit sample.txt
student@ubantu:~/OS$ ls
sample.txt
student@ubantu:~/OS$
```

25. cd ..

```
student@ubantu:~/new_folder$ pwd
/home/student/new_folder
student@ubantu:~/new_folder$ cd ..
student@ubantu:~$ pwd
/home/student
student@ubantu:~$
```

26. cd

```
student@ubantu:~$ mkdir new_folder
student@ubantu:~$ cd new_folder
student@ubantu:~/new_folder$ pwd
/home/student/new_folder
student@ubantu:~/new_folder$
```

Conclusion:

In this experiment, we have got some understanding about the basic Linux commands and their usages in performing various operations on the computer. We learned how to work with files and directories, etc.

Assessment Schemes: